3 Fig. 3.1 shows a cannon ball fired horizontally from a cannon at the edge of a cliff. The cannon ball travels a horizontal distance of 240 m in 2.5 s before hitting the sea.

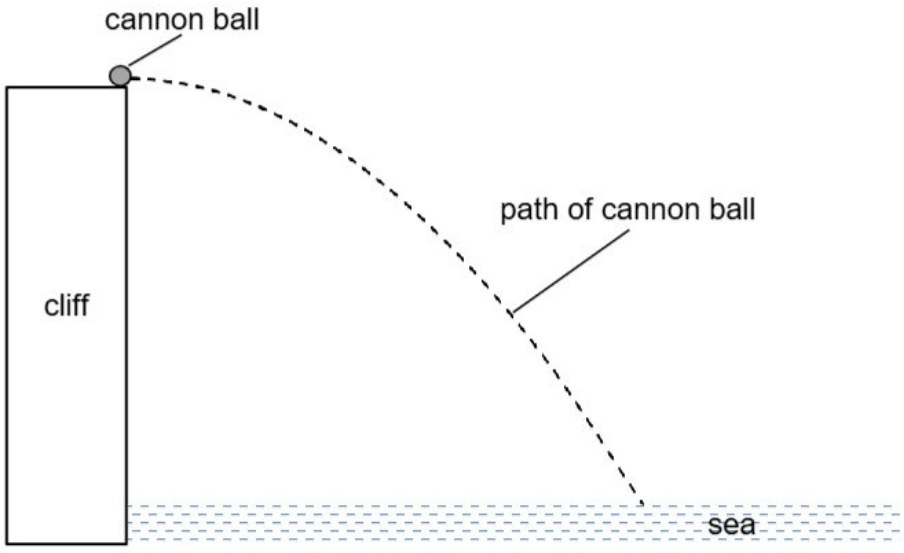


Fig. 3.1 (not to scale)

The effects of air resistance on the cannon ball are negligible.

(a) Calculate the height of cannon above sea level.

height = ... m [2]

(b) Determine the magnitude and direction of the velocity of the cannon ball just before striking the surface of the sea.

magnitude of velocity = ... $m\ s^{-1}$

direction: ..... [4]

(c) The initial speed of the cannon ball is now increased. Explain the effect that this will have on the time of flight of the cannon ball.

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..... [2]